
NAME

SDFFileUtil

SYNOPSIS

```
use SDFFileUtil ;

use SDFFileUtil qw(:all);
```

DESCRIPTION

SDFFileUtil module provides the following functions:

GenerateCmpdAtomAliasPropertyLines, GenerateCmpdAtomLine, GenerateCmpdBondLine, GenerateCmpdChargePropertyLines, GenerateCmpdCommentsLine, GenerateCmpdCountsLine, GenerateCmpdDataHeaderLabelsAndValuesLines, GenerateCmpdIsotopePropertyLines, GenerateCmpdMiscInfoLine, GenerateCmpdMolNameLine, GenerateCmpdRadicalPropertyLines, GenerateEmptyCtabBlockLines, GenerateMiscLineDateStamp, GetAllAndCommonCmpdDataHeaderLabels, GetCmpdDataHeaderLabels, GetCmpdDataHeaderLabelsAndValues, GetCmpdFragments, GetCtabLinesCount, GetInvalidAtomNumbers, GetUnknownAtoms, InternalBondOrderToMDLBondType, InternalBondStereochemistryToMDLBondStereo, InternalChargeToMDLCharge, InternalSpinMultiplicityToMDLRadical, IsCmpd2D, IsCmpd3D, MDLBondStereoToInternalBondStereochemistry, MDLBondTypeToInternalBondOrder, MDLChargeToInternalCharge, MDLRadicalToInternalSpinMultiplicity, ParseCmpdAtomAliasPropertyLine, ParseCmpdAtomLine, ParseCmpdBondLine, ParseCmpdChargePropertyLine, ParseCmpdCommentsLine, ParseCmpdCountsLine, ParseCmpdIsotopePropertyLine, ParseCmpdMiscInfoLine, ParseCmpdMolNameLine, ParseCmpdRadicalPropertyLine, ReadCmpdString, RemoveCmpdDataHeaderLabelAndValue, WashCmpd

METHODS**GenerateCmpdAtomAliasPropertyLines**

```
@Lines = GenerateCmpdAtomAliasPropertyLines($AliasValuePairsRef);
```

Returns a formatted atom alias property lines corresponding to successive pairs of atom number and alias values specified by a reference to an array. Two lines are generated for each atom number and alias value pairs: First line - A <AtomNum>; Second line: <AtomAlias>.

GenerateCmpdAtomLine

```
$Line = GenerateCmpdAtomLine($AtomSymbol, $AtomX, $AtomY,
                             $AtomZ, [$MassDifference, $Charge, $StereoParity]);
```

Returns a formatted atom data line containing all the input values.

GenerateCmpdBondLine

```
$Line = GenerateCmpdBondLine($FirstAtomNum, $SecondAtomNum,
                             $BondType, [$BondStereo]);
```

Returns a formatted bond data line containing all the input values.

GenerateCmpdChargePropertyLines

```
@Lines = GenerateCmpdChargePropertyLines($ChargeValuePairsRef);
```

Returns a formatted M CHG property lines corresponding to successive pairs of atom number and charge values specified by a reference to an array.

GenerateCmpdCommentsLine

```
$Line = GenerateCmpdCommentsLine($Comments);
```

Returns a formatted comments data line.

GenerateCmpdCountsLine

```
$Line = GenerateCmpdCountsLine($AtomCount, $BondCount,
                               $ChiralFlag, [$PropertyCount, $Version]);
```

Returns a formatted line containing all the input values. The default values of 999 and V2000 are used for

PropertyCount and Version.

GenerateCmpdDataHeaderLabelsAndValuesLines

```
@Lines = GenerateCmpdDataHeaderLabelsAndValuesLines(  
    $DataHeaderLabelsRef, $DataHeaderLabelsAndValuesRef,  
    [$SortDataLabels]);
```

Returns formatted data lines containing header label and values lines corresponding to all data header labels in array reference *DataHeaderLabelsRef* with values in hash reference *DataHeaderLabelsAndValuesRef*. By default, data header labels are not sorted and correspond to the label order in array reference *DataHeaderLabelsRef*.

GenerateCmpdIsotopePropertyLines

```
@Lines = GenerateCmpdIsotopePropertyLines($IsotopeValuePairsRef);
```

Returns a formatted M ISO property lines corresponding to successive pairs of atom number and isotope values specified by a reference to an array.

GenerateCmpdMiscInfoLine

```
$Line = GenerateCmpdMiscInfoLine([$ProgramName, $UserInitial,  
    $Code]);
```

Returns a formatted line containing specified user initial, program name, date and code. Default values are: *ProgramName* - *MayaChem*; *UserInitial* - *NULL*; *Code* - *2D*.

GenerateCmpdMolNameLine

```
$Line = GenerateCmpdMolNameLine($MolName);
```

Returns a formatted molecule name data line.

GenerateCmpdRadicalPropertyLines

```
@Lines = GenerateCmpdRadicalPropertyLines($RadicalValuePairsRef);
```

Returns a formatted M CHG property lines corresponding to successive pairs of atom number and multiplicity values specified by a reference to an array.

GenerateEmptyCtabBlockLines

```
$Lines = GenerateCmpdMiscInfoLine([$Date]);
```

Returns formatted lines representing empty CTAB block.

GenerateMiscLineDateStamp

```
$Line = GenerateMiscLineDateStamp();
```

Returns date stamp for misc line.

GetAllAndCommonCmpdDataHeaderLabels

```
($CmpdCount, $DataFieldLabelsArrayRef,  
    $CommonDataFieldLabelsArrayRef) =  
    GetAllAndCommonCmpdDataHeaderLabels(\*SDFILE);
```

Returns number of compounds, a reference to an array containing all unique data header label and a reference to an array containing common data field labels for all compounds in SD file.

GetCmpdDataHeaderLabels

```
(@Labels) = GetCmpdDataHeaderLabels(\@CmpdLines);
```

Returns an array containing data header labels for a compound

GetCmpdDataHeaderLabelsAndValues

```
(%DataValues) = GetCmpdDataHeaderLabelsAndValues(\@CmpdLines);
```

Returns a hash conating data header labes and values for a compound.

GetCmpdFragments

```
($FragmentCount, $FragmentString) = GetCmpdFragments(\@CmpdLines);
```

Figures out the number of disconnected fragments and return their values along with the atom numbers in a string delimited by new line character. Fragment data in FragmentString is sorted on based on its size.

GetCtabLinesCount

```
$CtabLinesCount = GetCtabLinesCount(\@CmpdLines);
```

Returns number of lines present between the 4th line and the line containing "M END".

GetInvalidAtomNumbers

```
($InvalidAtomNumbersCount, $InvalidAtomNumbers, $InvalidAtomNumberLines) =  
  GetInvalidAtomNumbers(\@CmpdLines);
```

Returns a list of values containing information about invalid atom numbers present in block or atom property lines.

GetUnknownAtoms

```
($UnknownAtomCount, $UnknownAtoms, $UnknownAtomLines) =  
  GetUnknownAtoms(\@CmpdLines);
```

Returns a list of values containing information about atoms which contain special element symbols not present in the periodic table.

InternalBondOrderToMDLBondType

```
$MDLBondType = InternalBondOrderToMDLBondType($InternalBondOrder);
```

Returns value of *MDLBondType* corresponding to *InternalBondOrder*.

InternalBondOrder	MDLBondType
-------------------	-------------

1	1
2	2
3	3
1.5	4

InternalBondStereochemistryToMDLBondStereo

```
$MDLBondStereo = InternalBondStereochemistryToMDLBondStereo(  
  $InternalBondStereo);
```

Returns value of *MDLBondStereo* corresponding to *InternalBondStereo* using following mapping:

InternalBondStereo	MDLBondStereo
--------------------	---------------

Up	1
UpOrDown	4
Down	6
CisOrTrans	3
Other	0

InternalChargeToMDLCharge

```
$MDLCharge = InternalChargeToMDLCharge($InternalCharge);
```

Returns value of *MDLCharge* corresponding to *InternalCharge* using following mapping:

InternalCharge	MDLCharge
----------------	-----------

3	1
2	2
1	3

-1	5
-2	6
-3	7

InternalSpinMultiplicityToMDLRadical

```
$MDLRadical = InternalSpinMultiplicityToMDLRadical(  
    $InternalSpinMultiplicity);
```

Returns value of *MDLRadical* corresponding to *InternalSpinMultiplicity*. These value are equivalent.

MDLBondStereoToInternalBondType

```
$InternalBondType = MDLBondStereoToInternalBondType($MDLBondStereo);
```

Returns value of *InternalBondType* corresponding to *MDLBondStereo* using mapping shown for *InternalBondTypeToMDLBondStereo* function.

IsCmpd2D

```
$Status = IsCmpd2D();
```

Returns 1 or 0 based on whether z-coordinate of any atom is non-zero.

IsCmpd3D

```
$Status = IsCmpd3D();
```

Returns 1 or 0 based on whether z-coordinate of any atom is non-zero.

MDLBondStereoToInternalBondStereochemistry

```
$InternalBondStereo = MDLBondStereoToInternalBondStereochemistry(  
    $MDLBondStereo);
```

Returns value of *InternalBondStereo* corresponding to *MDLBondStereo* using mapping shown for *InternalBondStereochemistryToMDLBondStereo* function.

MDLBondTypeToInternalBondOrder

```
$InternalBondOrder = MDLBondTypeToInternalBondOrder($MDLBondType);
```

Returns value of *InternalBondOrder* corresponding to *MDLBondType* using mapping shown for *InternalBondOrderToMDLBondType* function.

MDLChargeToInternalCharge

```
$InternalCharge = MDLChargeToInternalCharge($MDLCharge);
```

Returns value of *InternalCharge* corresponding to *MDLCharge* using mapping shown for *InternalChargeToMDLCharge* function.

MDLRadicalToInternalSpinMultiplicity

```
$InternalSpinMultiplicity = MDLRadicalToInternalSpinMultiplicity(  
    $MDLRadical);
```

Returns value of *InternalSpinMultiplicity* corresponding to *MDLRadical*. These value are equivalent.

ParseCmpdAtomAliasPropertyLine

```
@AtomNumAndValuePairs = ParseCmpdAtomAliasPropertyLine(  
    $CurrentLine, $NexLine);
```

Parses atom alias property lines in CTAB generic properties block and returns an array with successive pairs of values corresponding to atom number and its alias.

ParseCmpdAtomLine

```
($AtomSymbol, $AtomX, $AtomY, $AtomZ, $MassDifference, $Charge,  
    $StereoParity) = ParseCmpdAtomLine($AtomDataLine);
```

Parses compound data line containing atom information and returns a list of values.

ParseCmpdBondLine

```
($FirstAtomNum, $SecondAtomNum, $BondType) =  
  ParseCmpdBondLine($BondDataLine);
```

Parses compound data line containing bond information and returns a list of values.

ParseCmpdCommentsLine

```
$Comments = ParseCmpdCommentsLine($CommentsDataLine);
```

Returns the comment string.

ParseCmpdChargePropertyLine

```
@AtomNumAndValuePairs = ParseCmpdChargePropertyLine(  
  $ChargeDataLine);
```

Parses charge property line in CTAB generic properties block and returns an array with successive pairs of values corresponding to atom number and its charge.

ParseCmpdCountsLine

```
($AtomCount, $BondCount, $ChiralFlag, $PropertyCount, $Version) =  
  ParseCmpdCountsLine(@CountDataLines);
```

Returns a list of values containing count information.

ParseCmpdMiscInfoLine

```
($UserInitial, $ProgramName, $Date, $Code, $ScalingFactor1, $ScalingFactor2,  
  $Energy, $RegistryNum) = ParseCmpdMiscInfoLine($Line);
```

Returns a list of values containing miscellaneous information.

ParseCmpdIsotopePropertyLine

```
@AtomNumAndValuePairs = ParseCmpdIsotopePropertyLine(  
  $IsotopeDataLine);
```

Parses isotopic property line in CTAB generic properties block and returns an array with successive pairs of values corresponding to atom number and absolute mass of atom isotope.

ParseCmpdMolNameLine

```
$MolName = ParseCmpdMolNameLine($Line);
```

Returns a string containing molecule name.

ParseCmpdRadicalPropertyLine

```
@AtomNumAndValuePairs = ParseCmpdRadicalPropertyLine(  
  $RadicalDataLine);
```

Parses radical property line in CTAB generic properties block and returns an array with successive pairs of values corresponding to atom number and radical number value.

RemoveCmpdDataHeaderLabelAndValue

```
$NewCmpdString = RemoveCmpdDataHeaderLabelAndValue($CmpdString,  
  $DataHeaderLabel);
```

Returns a NewCmpdString after removing *DataHeaderLabel* along with its value from *CmpdString*.

ReadCmpdString

```
$CmpdString = ReadCmpdString(\*SDFILEHANDLE);
```

Returns a string containing all the data lines for the next available compound in an already open file indicated by SDFILEHANDLE. A NULL string is returned on EOF.

WashCmpd

```
($FragmentCount, $Fragments, $WashedCmpdString) =  
  WashCmpd(\@CmpdLines);
```

Figures out the number of disconnected fragments and return their values along with the atom numbers in a string delimited by new line character. Fragment data in FragmentString is sorted on based on its size.

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SEE ALSO

TextUtil.pm

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